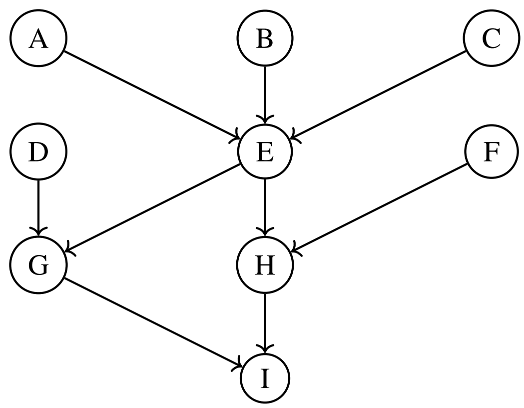
50.007 Machine Learning, Spring 2025
Homework 4

Due on April 21, 2025, 11.59 pm

In this homework, we explore Bayesian Networks. The network structure is shown below. All nodes take two values: $\{1, 2\}$.



Question 1 [15 points]

(a) Without knowing the actual value of any node, are node A and node F independent of each other?
(b) What if we know the value of node C and node I?
(c) Further, analyze the cases when only node C is observed and when only node I is observed. Provide a justification using the Bayes' ball algorithm.

Question 2 [15 points]

(a) Derive the formula for the *effective number of parameters* for a node i given its state space r_i (i.e., the number of possible values it takes) and the state spaces of its parents $\prod_{j \in \mathrm{pa}_i} r_j$.
(b) Compute the effective number of parameters needed for this Bayesian network when all nodes are binary (i.e., taking values $\{1, 2\}$).
(c) What would be the number of parameters if node D and node F can take 5 different values: $\{1, 2, 3, 4, 5\}$, and all other nodes can only take 3 different values: $\{1, 3, 5\}$? Explain how the increased state space affects the model complexity.

Question 3 [10 points]

(a) Given the probability tables for the nodes as provided below, compute $P(E = 2 \mid C = 1)$, clearly writing down all necessary steps.
(b) Now compute $P(E = 2 \mid C = 1, B = 2)$. Discuss how the additional observation affects the inference, and explain which conditional independencies simplify the calculation.

Probability Tables:

A	1	2
	0.2	0.8

B	1	2
	0.7	0.3

C	1	2
	0.2	0.8

D	1	2
	0.5	0.5

A	**B**	**C**	$E = 1$	$E = 2$
1	1	1	0.1	0.9
1	1	2	0.3	0.7
1	2	1	0.5	0.5
1	2	2	0.1	0.9
2	1	1	0.9	0.1
2	1	2	0.4	0.6
2	2	1	0.5	0.5
2	2	2	0.4	0.6

F	1	2
	0.3	0.7

D	**E**	$G = 1$	$G = 2$
1	1	0.1	0.9
1	2	0.6	0.4
2	1	0.6	0.4
2	2	0.5	0.5

E	**F**	$H = 1$	$H = 2$
1	1	0.1	0.9
1	2	0.4	0.6
2	1	0.5	0.5
2	2	0.5	0.5

G	**H**	$I = 1$	$I = 2$
1	1	0.1	0.9
1	2	0.9	0.1
2	1	0.7	0.3
2	2	0.9	0.1

Question 4 [10 points]

Assume that we do not have any prior knowledge about the probability tables for the nodes in the network, but we have the following 12 complete observations. Find a way to estimate the probability table for node A and for node H.

Observations:

Each row represents one sample (9 variables: $A, B, C, D, E, F, G, H, I$):

A	B	C	D	E	F	G	H	I
1	1	2	2	2	1	1	1	1
1	2	1	1	2	1	1	1	2
2	2	2	1	2	2	1	2	1
1	1	2	1	2	1	1	2	2
1	2	1	1	1	1	2	1	1
2	2	1	2	1	2	2	1	2
2	1	2	2	1	2	2	2	1
2	2	2	1	2	1	2	2	2
1	1	1	1	2	2	1	1	1
1	1	1	1	2	1	1	1	2
1	2	1	2	2	1	2	1	2
2	2	1	2	1	2	2	1	1

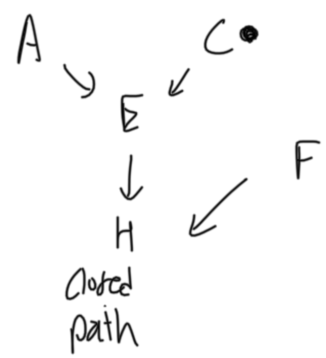
1. a) Path: $A \rightarrow E \rightarrow H \leftarrow F$

A, F are independent

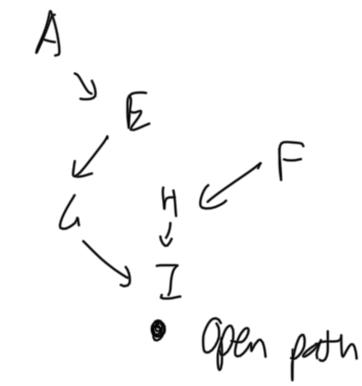
b) Path: $A \rightarrow E \rightarrow G \rightarrow I \leftarrow H \leftarrow F$

If I is known, path is open. Hence A and F are dependent given node C and I.

c) Only C is observed:



Only I is observed:



$\therefore$ A and F are dependent if only I is observed.

2. a) $\prod_{j \in Pa_i} r_j \cdot (r_i - 1)$

b) $r_i = 2$

No. of parameters for node $i = \prod_{j \in Pa_i} (2) \cdot (r_i - 1)$

$$= \prod_{j \in Pa_i} 2 = 2^{Pa_i}$$

Total parameters for i nodes $= \sum_i 2^{Pa_i} = \underset{A,B,C}{3 \times 2^0} + 2^3 + \underset{D,F}{2 \times 2^0} + 3 \times 2^2 = 25$

c) Number of parameters $= [(3-1) \cdot 1] \times 3 + [(5-1) \cdot 1] \times 2$

$$+ \underset{E}{[(3-1) \cdot 27]} + \underset{H}{[(3-1) \cdot 9]} + \underset{G}{[(3-1) \cdot 9]}$$

$$+ \underset{I}{[(3-1) \cdot 9]} = 122$$

Model complexity increases as the number of parent configurations and parameters increase with increased state space, resulting in more memory used for CPTs and more computation.

3. a) $P(E=2 \mid C=1) = \dfrac{P(E=2, C=1)}{P(C=1)}$

$$P(E=2, C=1) = \sum_a \sum_b P(E=2, C=1, A=a, B=b)$$

$$= P(E=2 \mid A=a, B=b, C=1) \cdot P(A=a) P(B=b) P(C=1)$$

$$\frac{P(E=2, C=1)}{P(C=1)} = \sum_{a,b} P(E=2 \mid A=a, B=b, C=1) \times P(A=a) \times P(B=b)$$

$$= (0.9 \times 0.2 \times 0.7) + (0.5 \times 0.2 \times 0.3) + (0.1 \times 0.8 \times 0.7) + (0.5 \times 0.8 \times 0.3)$$

$$= 0.332$$

b) $P(E=2 \mid C=1, B=2) = \dfrac{P(E=2, C=1, B=2)}{P(C=1, B=2)}$

$$P(E=2, C=1, B=2) = \sum_a P(E=2 \mid A=a, B=2, C=1) \cdot P(A=a) \cdot P(B=2) \cdot P(C=1)$$

$$\frac{P(E=2, C=1, B=2)}{P(C=1, B=2)} = \sum_a P(E=2 \mid A=a, B=2, C=1) \cdot P(A=a)$$

$$= 0.5 \times 0.2 + 0.5 \times 0.8$$

$$= 0.5$$

Conditional independence between A, B simplifies the calculation. The additional observation B=2 in b) reduces the complexity of the inference as B now no longer needs to be summed over.

4.

A	1	2
	$7/12$	$5/12$

H

E	F	1	2
1	1	$1/1$	0
1	2	$2/3$	$1/3$
2	1	$4/6$	$2/6$
2	2	$1/2$	$1/2$